

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 2: Compound Interest (Without Using Formula)

Every formula and result of the chapter on one sheet

|    |                                |                                                                                                    |
|----|--------------------------------|----------------------------------------------------------------------------------------------------|
| 1  | Simple interest                | $\frac{P \times R \times T}{100}$                                                                  |
| 2  | Amount                         | $P + \text{interest}$                                                                              |
| 3  | Interest for one year          | $\frac{P \times R}{100}$                                                                           |
| 4  | Principal for the next year    | Amount at the end of this year                                                                     |
| 5  | Compound interest              | Final amount – original principal                                                                  |
| 6  | First year                     | C.I. = S.I.                                                                                        |
| 7  | Rate from successive interests | $\frac{\text{difference of C.I. of successive years}}{\text{C.I. of the earlier year}} \times 100$ |
| 8  | Rate from successive amounts   | $\frac{\text{difference of the amounts}}{\text{earlier amount}} \times 100$                        |
| 9  | C.I. – S.I. for 2 years        | Interest for one year on the first year's interest                                                 |
| 10 | Half-yearly compounding        | Rate $\frac{R}{2}\%$ per half-year, $2T$ periods                                                   |
| 11 | After a repayment              | New principal = amount – repayment                                                                 |
| 12 | Depreciation                   | New value = old value – year's decrease                                                            |